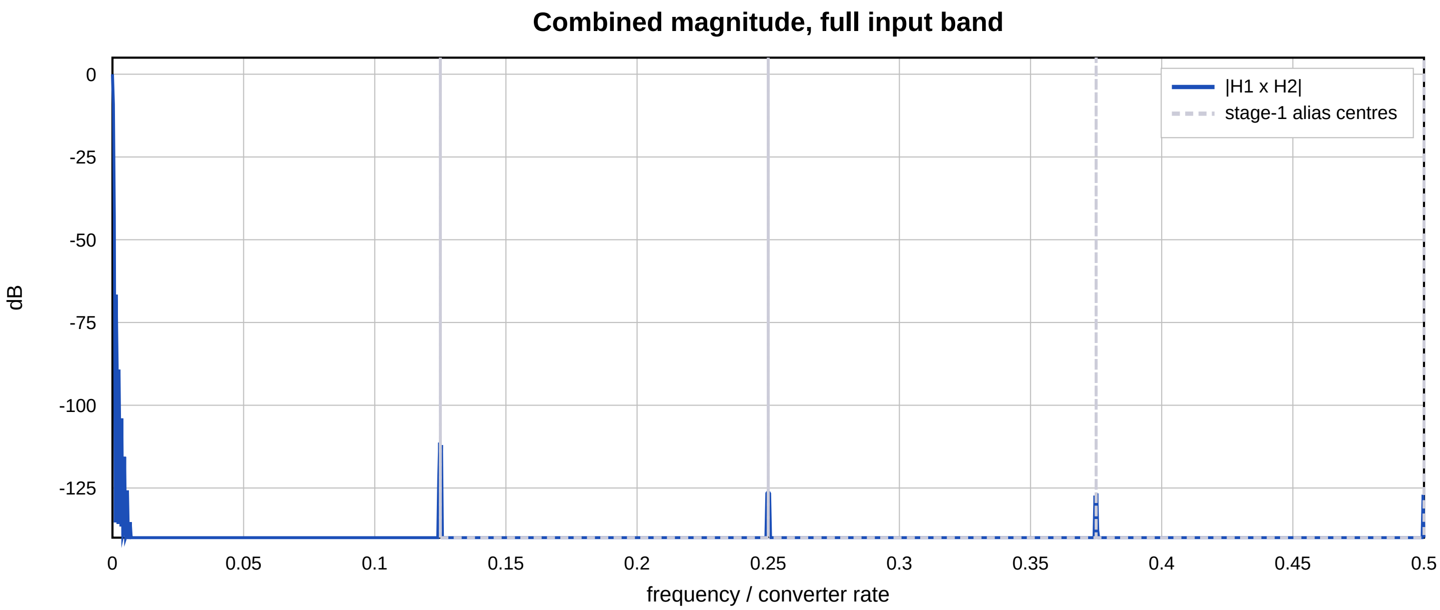
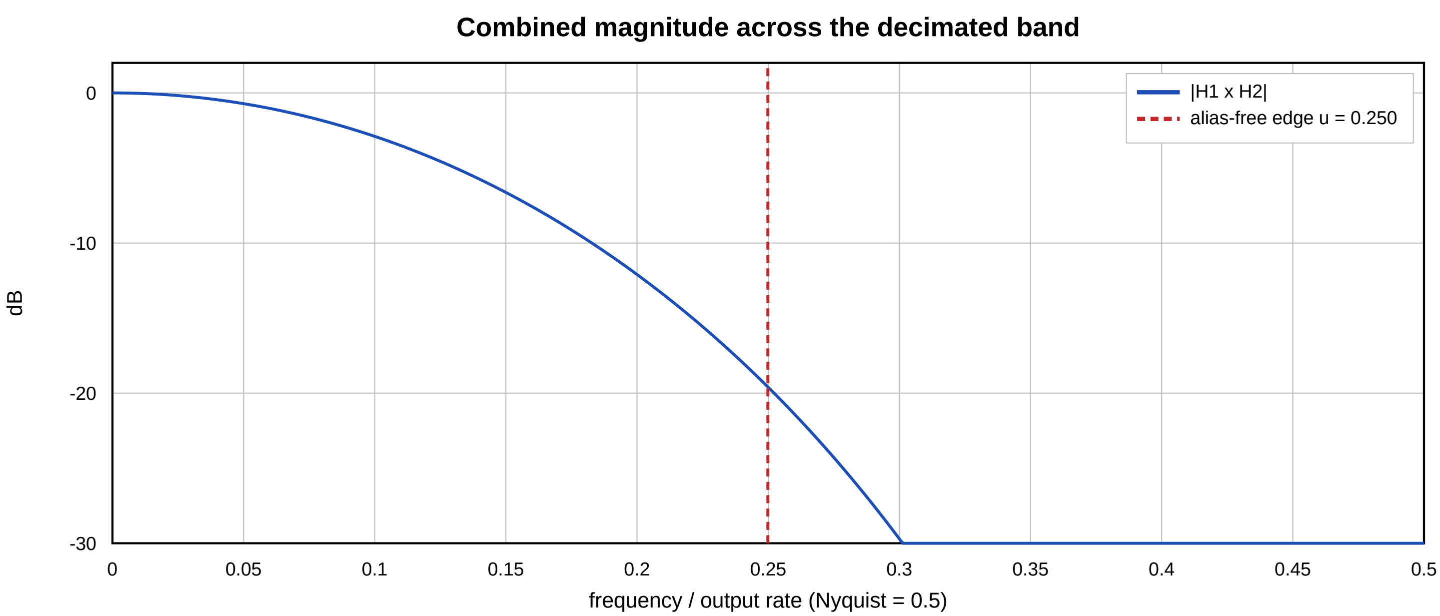
Decimation Chain - Derived Design

125.000 MHz / 488 = 256.1475 kHz, split [8, 61], 64.000 kHz alias-free





Stages

stage 1: /8 N=2 M=1 at 125.000 MHz
accumulator 22 bits, prune budget 12, widths [16, 16, 16, 16]
64 register bits, in 16 bits out 16 bits

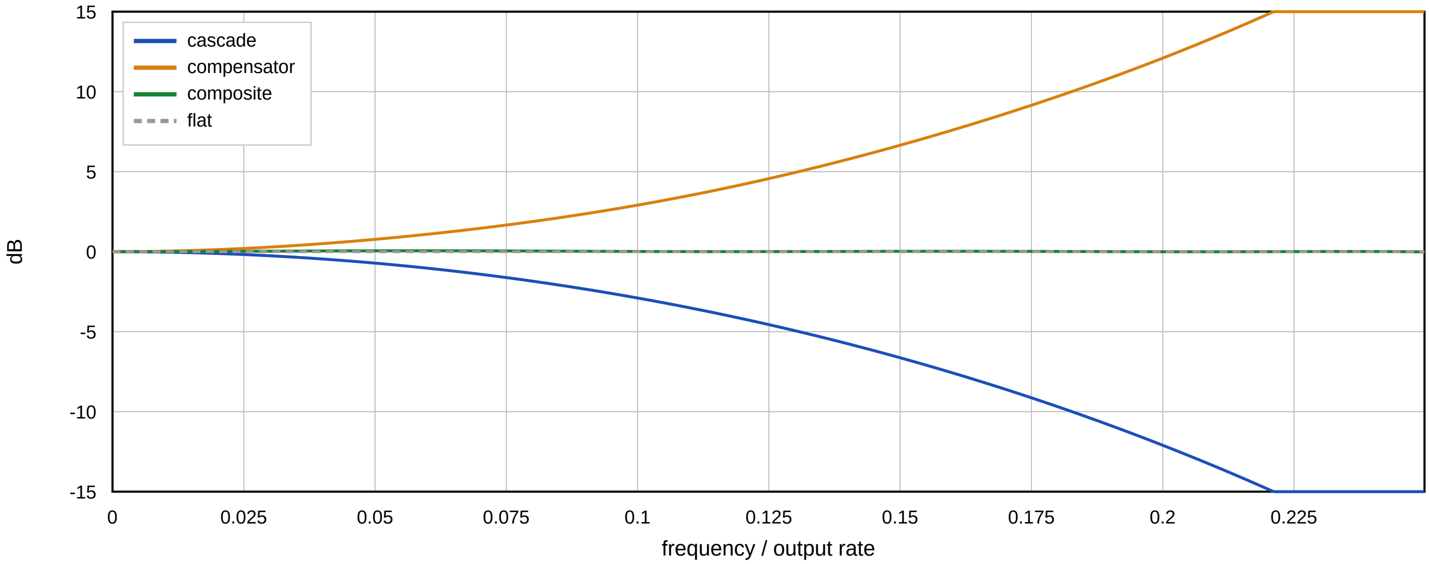
stage 2: /61 N=5 M=2 at 15.625 MHz
accumulator 51 bits, prune budget 47, widths [37, 30, 24, 18, 16, 16, 16, 16, 16, 16]
205 register bits, in 16 bits out 16 bits

A stage's nulls must reject the aliases of its own decimation: once energy has folded into the band no later stage can remove it, so every stage carries the full requirement.

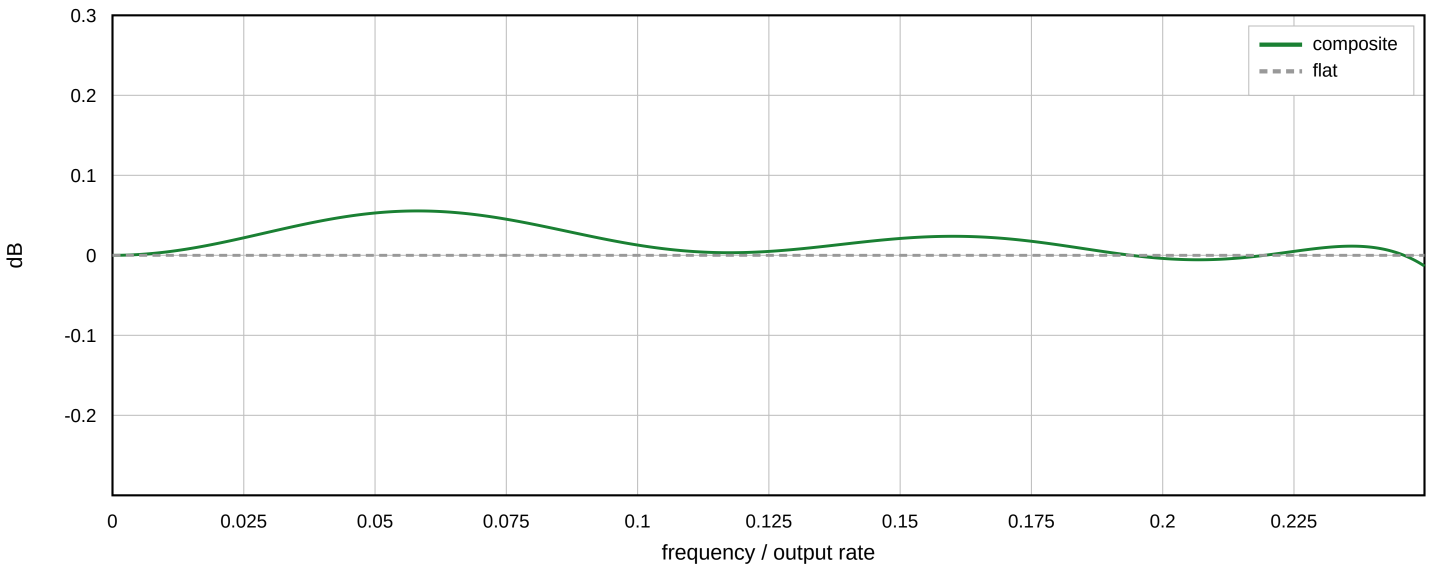
Compensation and Result

11 taps, 16-bit coefficients, 6 multipliers

Cascade, compensator, and the two together



Composite, zoomed - what is left



Achieved

ripple .... 0.0689 dB (asked <= 0.100)
alias rejection ... 67.3 dB   (asked >= 60.0)
output SNR .... 84.6 dB   (asked >= 80.0)

group delay .... 6862 input samples = 54.9 us
  cascade 2427, integrator pipeline 33, comb pipeline 1960, output regs 2, compensator 2440
  largest term is the compensator at 2440 (36%); loop bandwidth ~ 1.8 kHz

register bits ..... 269 (rate-weighted cost 89.6)
taps .... [-177, 1286, -4829, 12002, -21186, 26320, -21186, 12002, -4829, 1286, -177]
DC gain .... 1.000000 (exact by construction)
alternative .... [4, 61, 2], cost 96.8, 384 bits - feasible, but costlier by the rate-weighted model

Rate-weighted cost is a proxy, not an area figure: flops cost the same however slowly
they are clocked. It captures that a wide adder at the converter rate is the hard part.